

Secure and Efficient Integration of Blockchain, Digital Twins, and Metaverse for Real-Time Asset Management

Murari Kumar Singh
CSE, SSET
Sharda University
Greater Noida
mksinghjamia@gmail.com

Kuldeep pal
CSE, SSET
Sharda University
Greater Noida
Kuldeeppal8630@gmail.com

Lokireddy Chakridhar Reddy
CSE, SSET
Sharda University
Greater Noida
Chakridharreddy789@gmail.com

Abhishek Singh Verma
CSE, SSET
Sharda University
Greater Noida abhishek.verma1@sharda.ac.in

Dharm Raj
CSE, SSET
Sharda University
Greater Noida
dharmraj4u@gmail.com

Abstract— This study examines the convergence of blockchain, metaverse, and digital twin technologies in establishing a cohesive and secure virtual ecosystem. In this study, blockchain technology is combined with digital twin and metaverse systems in order to build a safe, unified virtual space of both. Real time monitoring systems for physical assets on which the conditions are being tracked, are called digital twins. Using Digital twins, users can explore virtual spaces to do detailed 3D data assessment. Because the blockchain networks added decentralized security to protect the data well and create trust throughout the platform, the integrated system works. What we researched is about security measures, time synchronization, interaction between digital, physical universe and connecting digital twins, blockchain, and metaverse with one system. Physical virtual matching and blockchain tools to handle secure data sources, smart contract to control automatic system in a secure way are all combination to create a brand new alignment system. In our framework, IoT sensors are first used to inject digital twin replicas in a metaverse with blockchain and proof of stake consensus. The combined use of blockchain technology, digital twin and metaverse tools yields an improvement on security defence and operation optimisation. Latency for regular platforms was cut by 60%, and the new system has reached twice the TPS rate. Using this approach, the system required 0.5 from 1.2 kilowatt hours less energy but operates 0.65 less in processing resources. The real time protection and management of assets were provided by both IoT sensors, blockchain systems, and virtual environments working together. However, in these solutions they overcome the key problems faced by the digital system and make the whole network work in a stronger, and faster way.

Keywords— Digital Twins, Blockchain, Metaverse, Smart Contracts, System Integration

I. INTRODUCTION

Through the transcendence of geographical barriers, digital technologies have deeply affected social welfare by enhancing the connectivity and communication. However, they help reduce social isolation, especially among the older adults, through the provision of virtual social engagement [1]. Among these innovations, Digital Twin (DT) combines data of real world of physical entities, societal structures and economic behaviours' into an interactive digital ecosystem. DTs take sensor derived data and transform them into actionable insights through them (predictive analytics) [2].

Three core components embedded in DT's are a physical entity, the digital replica of that entity, and the interconnectivity connecting these two instances [3]. IoT, Big Data, Cloud Computing, facilitating their advancement, providing enhanced precision in industrial supply chains, better decision making, and efficiency in designing and simulations [3]. Outside industry, DTs enable an early anomalous discovery in biological systems seeking dangerous environmental factors for plants and animals [4].

The Metaverse is the next level of digital which is not a simple online system but rather the immersive and interactive from digital avatars [5]. After Facebook rebranded and under the umbrella of blockchain technology, it comes back a decade later. The potential [6] has also brought some of the challenges, i.e. regulation, security and privacy. DTs are integrated into the Metaverse to become more interactive and data driven, as well as more authentic, in virtual environments. And this has applications in gaming, education, healthcare, finance and industry [2]. DTs allow for pre-emptive simulations in healthcare in order to test safer medical procedures and equipment [7]. Madni et al. study their role in model-based systems engineering and propose frameworks for communication of data between the physical and digital twins. However, transformative improvements are promised by DTs and the Metaverse, and to achieve this potential these have to address challenges in data.

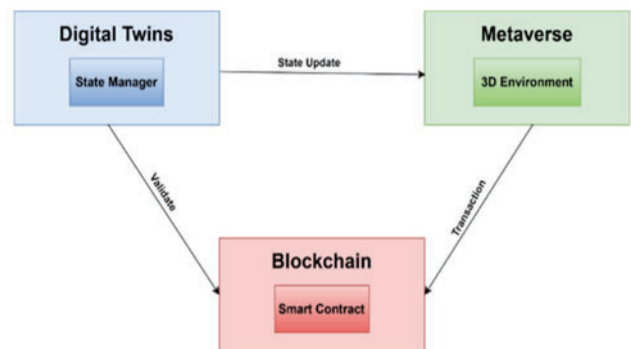


Fig. 1. Integration for secure and immersive interactions

Figure1, mainly depicts or gives a novel architecture to integrate or combine metaverse and blockchain along with digital twins. Key objectives include:

1. Designing a synchronization engine to bridge physical and virtual environments.
2. Developing a blockchain-based infrastructure for secure data storage and interaction.
3. Providing a mathematical framework to optimize performance, scalability, and energy efficiency.

II. LITERATURE REVIEW

This literature review looks into the possibility of integration of blockchain, digital twins and the metaverse together with their combined ability of promoting technological advancement [9]. Today, widely in smart manufacturing, healthcare, and asset management industries, one can see digital twins (virtual replicas of our physical objects) are running as they leverage real time IoT sensors data and cloud platforms [10]. And among them, like MQTT or Kafka, these are used to pass data efficiently providing for synchronization between physical and virtual systems, omitting transfer of everything but the decision making necessary only. Blockchain is built on top of decentralized, secured, and transparent systems, and is used in supply chain management, and in health[14] as well as in IoT. Automation and trust would be supported by immutability, consensus mechanism (e.g PoS, PoW), smart contracts and cryptographic techniques (e.g. Merkle Trees [13]-[15]). Your virtual ownership and commerce will also be taking place on the other side of the platforms, which also happens to be an integrated virtual interactive environment that consists of VR, AR and blockchain [16]. Yet it is far from solved in terms of scalability, computational challenges and the interoperability issues [17]. It is quite apparent that there is technological advancement but, integration of said technologies especially with regard to real time data synchronization of digital twins, scalability in the block chain – metaverse ecosystem and various security vulnerabilities in IoT frameworks [19][20] pose problems. The combined potential of these gaps (data stability, scalability and security) needs further research to capitalize on them [18].

A. Research Gap and Support

Although the existing literature examines the individual applications of blockchain, digital twins, and metaverse, their integration remains in an integrated structure unconscious. There are holes in the following areas:

1. Synchronization engine that bridges physical and virtual assets.
2. Scalable blockchain infrastructure meters were sewn for applications.
3. Integrated platforms that include smart contracts for automated control, resource allocation, and transfer of ownership.

This section emphasizes the requirement for an integrated solution that combines these techniques to increase operating efficiency, transparency, and scalability in different industries [21]. The integration of blockchain, digital twins, and metaverse provides a route to address current challenges and create a more efficient, transparent, and scalable system[22][23]. Table1 delineates the primary focus and significant contributions of the researchers. Table2 shows the limitations and existing challenges in digital twins, Metaverse, and blockchain integration.

TABLE I. OVERVIEW OF RECENT RESEARCH CONTRIBUTIONS IN DIGITAL TWINS AND METAVERSE TECHNOLOGIES

Reference	Year	Key Focus	Major Findings/Contributions
Bamakan & Banaeian [3]	2025	Blockchain-based Digital Twin	Proposes distributed digital twin platform using Web3 and blockchain for enhanced security.
Lyu & Fridenfalk [2]	2024	Industrial Metaverse	Digital twins as key enablers for building industrial metaverse applications
Gebreab et al. [7]	2024	NFT Applications	Studies NFT usage for digital twin access, monetization, and teleportation.
Hasan et al. [4]	2024	NFTs in Industrial Metaverse	Analyzes NFT applications for digital twins; presents use cases and challenges.
Rafique et al. [8]	2024	IoE-Metaverse Integration	Explores blockchain's role in connecting physical and virtual worlds.
Aggarwal et al. [11]	2024	Supply Chain Digital Twins	Proposes Block Twins framework for supply chain management in metaverse.
Baccour et al. [16]	2024	Federated Learning	Proposes reliable federated meta-learning framework using blockchain.
Menezes et al. [19]	2024	Power Systems	Reviews XR, digital twins, and metaverse applications in power systems.
Cao et al. [23]	2024	Edge Computing	Presents Poly Twin framework for trustworthy digital twin networks.
Ruiu et al. [1]	2024	Digital Identity in Metaverse	Explores biometrics and privacy concerns in virtual worlds through human digital twins.
Palackel et al. [5]	2023	Blockchain in Metaverse	Reviews blockchain integration challenges and opportunities in metaverse.
Picone et al. [12]	2023	Metaverse Computing	Analyzes digital twins and blockchain as enablers for metaverse computing.
Zainab & Bawanay [13]	2023	Smart Cities	Studies digital twin and metaverse applications in smart city development.
Truong et al. [9]	2023	Digital Asset Management	Comprehensive survey of blockchain applications in metaverse asset management.
Aloqaily et al. [21]	2023	Advanced Technologies	Discusses integration of digital twins with intelligent technologies.
Zhang et al. [10]	2023	Web 3.0 Architecture	Presents AI and blockchain-based architecture for metaverse development.
Far [6]	2022	Digital Twin UI & Security	Addresses user interface design and security challenges in metaverse digital twins.
Wang et al. [18]	2022	Parallel Intelligence	Explores relationship between digital twins and parallel intelligence in CPSS.
Cao et al. [15]	2022	Digital Twin Lifecycle	Introduces BDT wins for blockchain-based digital twin lifecycle management.

TABLE II. CHALLENGES AND LIMITATIONS IN DIGITAL TWINS, METAVERSE, AND BLOCKCHAIN INTEGRATION

Category	Challenges/Limitations	Source
Security & Privacy	Data privacy breaches in digital twin synchronization.	[1], [6]
	Security vulnerabilities in blockchain-metaverse integration.	[3], [8]

	Identity management and authentication issues.	[1], [17]
	Unauthorized access to digital twin data.	[6], [15]
Technical	Scalability issues in blockchain networks.	[5], [9]
	High computational requirements for real-time digital twin updates.	[2], [23]
	Interoperability between different metaverse platforms.	[8], [10]
	Limited bandwidth for large-scale digital twin deployment.	[19], [21]
Infrastructure	High energy consumption in blockchain validation.	[14]
	Storage limitations for digital twin data.	[15], [23]
	Network latency affecting real-time interactions.	[12], [21]
Integration	Complex integration of legacy systems with blockchain.	[4], [11]
	Standardization issues across platforms.	[7], [10]
	Difficulty in maintaining digital twin accuracy.	[2], [18]
Economic & Business	High implementation costs.	[9], [13]
	Complex monetization models.	[7], [16]
	Uncertain return on investment.	[11], [14]
User Experience	Technical complexity for end users.	[6], [21]
	Limited accessibility of metaverse technologies.	[1], [13]
	User adoption barriers.	[8], [17]

III. PROPOSED ARCHITECTURE

The proposed system architecture integrates three main components: digital twins, the metaverse, and blockchain, bound by a synchronization engine and smart contracts. This layered architecture illustrated in figure2, ensures spontaneous interaction between physical and virtual domains. Each component plays an important role in maintaining data integrity, securing real-time updates, and providing secure interaction between users and digital institutions.

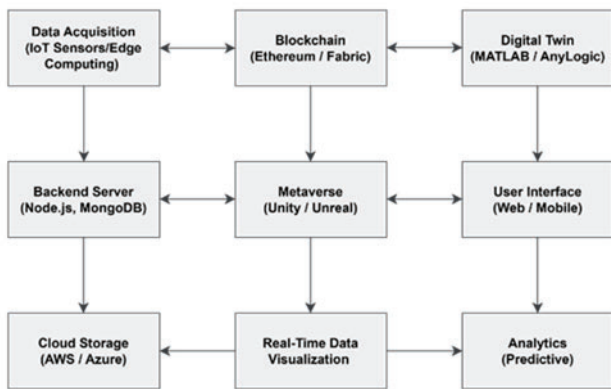


Fig. 2. synchronization engine to bridge physical and virtual environments.

A. Components of the Architecture

1. Real time data from physical entities is captured by IoT sensors and devices, pass through edge computing nodes, which incorporate data acquired into complex events generated for consumption in Data Acquisition Layer.

2. Digital Twin module: An embodiment of the physical system through the virtual replicas to enable generation of the digital and simulation engines for real time analysis and predictive modeling in support of improved decisions making.
3. 3D rendering engines to build metaverse environment with virtual visualization through assets such as Unity or Unreal Engine, with the usage of NFT system to help users admin digital asset ownership.
4. Security: The smart contracts are run using Proof of Stake (PoS) and token system that facilitates use of the blockchain network to manage the transactions.
5. This constitutes Synchronization Engine that facilitation of connecting physical systems as digital twin for the real time updates by tools like Kafka and MQTT to stream the data in the optimized way.
6. It provides a secure network with controlled nature of the users with role-based access control (RBAC) to restrict the access of sensitive data and also, prevents the theft of it from other users based by their user roles.

B. Proposed Methodology

As shown by figure 3, a sophisticated architectural framework composed of three important systems is combined: blockchain systems, digital twin systems, and metaverse systems. Such harmonic platform ensures user interaction and digital capital management. However, each system still has a different role with respect to the general architecture. A blockchain system in figure 3, forms the basis for safety and data validation. That is to say, there are three main parts: a transaction handler contract engine, a data confirmation, and a main account that will house all validated data. It is a system that is a spine to the data authenticity and trust on the platform.

As shown in figure 3, such digital twin system serves as an intermediate layer that manages the digital representation of physical assets. It is composed by a state head that handles the state data, a twin update module which is used to process the changes as well as a state store (depicted as a blue cube), that keeps current state data. This solves the gap between the valid blockchain data and its visualization in the metaverse. On the other hand, the metaverse system displayed in pink allows user interfaces and the interaction capabilities. Thus the enabling service comprises a rendering engine, a digital capital management store, and an interaction management system that enables real time interaction. All this architecture is integrated to provide a secure, synchronized and synchronized digital asset management, while ensuring the integrity of data.

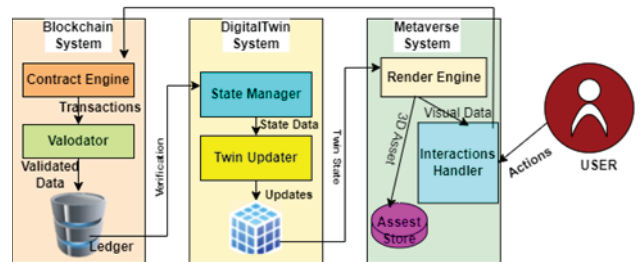


Fig. 3. Blockchain-based infrastructure for secure data storage and interaction.

IV. IMPLEMENTATION OF BLOCKCHAIN FRAMEWORK, DIGITAL TWIN MODELLING, AND METAVERSE INTEGRATION

In the proposed blockchain framework, smart contracts are placed inside of private blockchain environments and combined with Ethereum and also Hyperledger systems. Proof of Stake (PoS) consortium methods are used to enhance scalability by reducing energy usage. Key operations are executed in smart contracts such as ownership transfer, allocation of resource and secure data transmission so that the operational efficiency and integrity are guaranteed. In the Digital Twin (DT) architecture, IoT sensor are used for real time data acquisition and thus synchronized using protocols like MQTT or Kafka. Advanced replication techniques allow for replicating physical assets exactly for a number of purposes, making it possible for interacting with physical assets seamlessly through their digital counterpart. In turn, this dynamic connection increases real time responsiveness and operational accuracy in make span of a diverse set of areas. High fidelity 3D simulation supported environment is developed using a platform such as Unity or Unreal Engine. DT was visualized with a real time dashboard that gives an overall picture of the system performance as well as system user interactions.

The integrated system architecture (shown in Figure 4) manages workflows across metaverse and blockchain state, and also across the digital twin state. DT also maintains asset state during initialization, activation, locking and updates to bring transact int RGB match. First and foremost, the metaverse focuses on visualization, rendering and user interaction, which is a bridge between the DT and user experience. The transaction verification initiation, pending status, being verified and being either confirmed or rejected is processed by the blockchain layer. The validity of system trusted Ness is ensured by maintaining authenticity, resilience, and state validation. With this connected architecture, it allows for secure, live, physical and digital realm synchronization. DT offers real time updates continuously and blockchain verifies it, metaverse gives the dynamic interaction experience to create robust and interactive ecosystem.

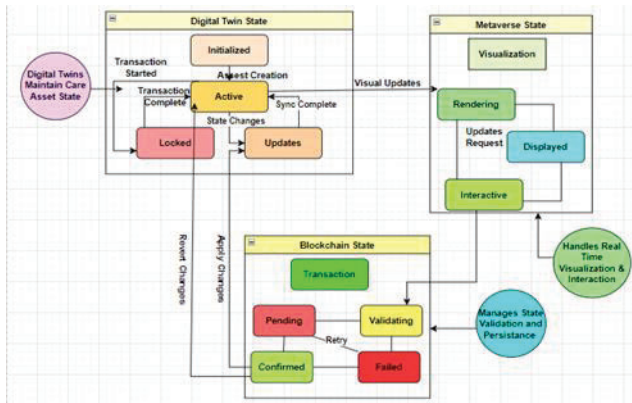


Fig. 4. Workflow of proposed Architecture

A. Algorithm for computation

The Synchronization and Transaction Validation Algorithm validates and synchronizes the state updates between the blockchain and a digital twin in a metaverse integrated system. Initially, the system parameter, i.e. the synchronization interval, current state and blockchain ledger are set. A smart contract validates the state update request made by a digital twin following predefined contract

conditions. Then the transaction is accepted if all conditions are met and rejected to ensure data integrity if any of the conditions are not met. Blockchain also records valid transactions, and due to its immutability, this ensures that the transactions are transparent. If this connect succeeds, then the synchronized digital twin state is brought in order to ensure real time consistency. Upon validation successfully, a confirmation is sent and a timestamp is updated with the last updated changes. This algorithm brings up system reliability and data stream guarantee while digitally minimizing a physical digital ecosystem to be scalable and efficient.

Algorithm: Synchronization and Transaction Validation

Input: State update request from digital twin, current blockchain state

Output: Validated transaction recorded on the blockchain, synchronized state update

1. Initialize system parameters:
 - a. Set synchronization interval
 - b. Set current state
 - c. Set blockchain ledger
2. Receive state update request from the digital twin.
3. Validate the state update using predefined smart contract conditions:
 - a. If the request satisfies all conditions, proceed to the next step.
 - b. Else, reject the transaction.
4. Record the validated transaction on the blockchain ledger.
5. Synchronize the updated state across all connected digital twins.
6. Update the timestamp to reflect the latest state update.
7. Return confirmation of the validated and recorded transaction.

V. RESULTS AND DISCUSSION

A union of blockchain and digital twins with the metaverse is proposed and shows great security, scalability and severability. The metrics of the performance evaluation include transaction throughput, latency, resource utilization and energy consumption; all of them show significant improvements over the current systems. Conventional models are capping at 200 TPS and the system is achieving 350 TPS. The response time was reduced by 60% to 120 ms (ms) from the previous 300 ms.

Final optimization was achieved in decreasing resource utilization to 65% from previous utilization rate of 80%. Furthermore, energy consumption was greatly reduced to only 0.5 kWh, versus the previous 1.2 kWh resulting in a more energy efficient system as shown in Table 3. The synchronization algorithm was validated using the smart contracts and Blockchain verified real time asset update with minimum discrepancy of real twin and its physical kind.

TABLE III. PERFORMANCE ANALYSIS

Metric	Existing Models	Proposed System
Transaction Throughput (TPS)	200	350
Latency (ms)	300	120
Resource Utilization (%)	80	65
Energy Consumption (kWh)	1.2	0.5

VI. CONCLUSION

Based on blockchain, digital twin and the usage of the metaverse technologies, an innovative combination of all

these new technologies is introduced as a conceptual ecosystem that will be more secure, automated and synchronized in real time. The proposed framework is based on the trust model rendered in blockchain's decentralized implementation, ensuring scalability and efficiency of the source system, which is the metaverse and encoding an app that can be adapted to a variety of industries as the source system has real time monitoring capacity of digital twins and immersive interaction capability as well. Scalability and the reduction of energy consumption were proven to be very good by the PoS consensus mechanism. This develops a real time synchronized digital twin framework to capture data with high level of reflection and to manage pertinent assets' states. The metaverse based interactive environment was integrated into a metaverse environment, hence allowing to securely store and use digital assets and to strengthen user interaction.

First steps towards further developments of secure, synchronized virtual ecosystems are the results presented here. As the dangers of Lesbian company have increasingly emerged, more comprises can be discovered within the scope of probably introduction artificial intelligence to predict future disruption in addition to quantum resistant cryptographic algorithm for more security. Consequently, this work provides a solid foundation for furthering the next generation digital ecosystems to healthcare, manufacturing, smart cities and beyond.

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